

ARISTO EDUCATIONAL PRESS
HONG KONG DIPLOMA OF SECONDARY
EDUCATION EXAMINATION

(Mock Test 27)

Suggested Solution

Paper 1

- | | | |
|-------|-------|-------|
| 1. B | 11. C | 21. D |
| 2. A | 12. A | 22. C |
| 3. C | 13. B | 23. A |
| 4. D | 14. C | 24. D |
| 5. A | 15. D | 25. D |
| 6. A | 16. D | 26. C |
| 7. B | 17. C | 27. D |
| 8. D | 18. D | 28. B |
| 9. B | 19. C | 29. C |
| 10. A | 20. C | 30. B |

Explanation

1. Answer: B

Karen's cost of taking a fine art programme in France is the value of teaching drawing in a gallery.

Option B: The gallery raising the salaries of all teachers would increase the value of teaching drawing in a gallery, thus increasing the cost of taking a fine art programme in France.

Option A: The option of teaching drawing in a gallery no longer exists. As there is no choice, there is no cost.

Options C and D: These would only affect the value of taking a fine art programme in France without affecting the value of teaching drawing in a gallery. Karen's cost of taking a fine art programme in France will not be affected.

2. Answer: A

Option B: Even if a good is scarce, if the market price is in equilibrium, the quantity demanded will be equal to the quantity supplied, and there will be no shortage of the good.

Option C: Scarcity refers to the situation in which the resources available are not enough to satisfy all people's wants, but it does not mean that everyone in the world wants to obtain the good.

Option D: Even if a good is scarce, it can be free of charge.

3. Answer: C

(1): The drinking water is produced from scarce resources. It is thus an economic good, not a free good.

(2), (3): Since the drinking water is used in the production process to produce another service (catering service), it is a producer good, not a consumer good.

(4): Since the drinking water is rival and excludable in consumption, it is a private good.

4. Answer: D

The unit is the private property of Mr. Li, so he has a full set of private property rights over it.

5. Answer: A

Before changing the firm's form of business ownership, the firm does not have an independent legal status, so the firm's form of business ownership could be sole proprietorship or partnership.

After changing the firm's form of business ownership, the firm has an independent legal status but does not need to disclose its financial status. The firm's form of business ownership is a private limited company.

6. Answer: A

The shares of a private limited company are not freely transferable while the shares of a public limited company are, so the risk of a private limited company being taken over is lower.

Option B: Whether a firm earns a higher rate of return has nothing to do with its form of business ownership.

Option C: Both a private limited company and a public limited company can issue shares and bonds.

Option D: Both a private limited company and a public limited company are subject to the same profits tax rate.

7. Answer: B

If one person uses a parking space, the number of parking spaces available to others will decrease. The parking spaces are rival in consumption and are therefore private goods.

8. Answer: D

(1): Since Mr. Wong is the owner of the company, he is the human effort that makes decisions and bears business risks, so he is an entrepreneurship. His workers are the human effort used in production, so they are labour.

(2): Different workers specialise in different stages of a renovation project, so the company adopts a complex division of labour.

(3): Since each worker specialises in a particular stage of a renovation project, they do not have to learn every skill. Training time can thus be reduced.

9. Answer: B

This means that people need to devote more training time and resources to these industries, resulting in an increase in the cost of changing occupations. This reduces the ability and willingness of workers to change from one industry to another.

Option A: This will not affect the ability and willingness of a worker to change from one occupation to another, so the occupational mobility of labour will not be affected.

Option C: This will increase the geographical mobility of labour.

Option D: This will increase the occupational mobility of labour.

10. Answer: A

Jimmy can earn a higher income by providing better services.

(2): Since the commission is not fixed, Jimmy's income will be less stable than that under the time rate scheme.

(3): This is an advantage of the arrangement from the viewpoint of the salon, not the hairstylist.

11. Answer: C

As there is a fixed factor (machines), the firm is operating in the short run.

Machines (units)	Labour (units)	Average product (units)	Total product (units)	Marginal product (units)
5	5	100	500	-
5	6	110	660	160
5	7	120	840	180
5	8	125	1,000	160

When labour increases from 7 units to 8 units, the marginal product of labour decreases from 180 units to 160 units, so the law of diminishing marginal returns applies.

12. Answer: A

(1), (3): These will increase with longer opening hours.

(2), (4): These are fixed costs for the beauty salon and will not change with the opening hours.

13. Answer: B

A horizontal integration does not diversify business, and thus cannot spread risk.

14. Answer: C

Output (units)	Average cost (\$)	Total cost (\$)	Marginal cost (\$)
1	15	15	-
2	16	32	17
3	17	51	19
4	18	72	21
5	19	95	23

When the market price decreases from \$21 to \$19, the firm's total cost decreases from \$72 to \$51. When the market price is \$21, the output is 4 units and the total revenue is \$84. Profit is \$12 ($= \$84 - \72). When the market price is \$19, the output is 3 units and the total revenue is \$57. Profit is \$6 ($= \$57 - \51). Therefore, the profit of the firm will decrease by \$6.

15. Answer: D

The law of demand states that an increase in the price of a good will result in a decrease in its quantity demanded, *ceteris paribus*.

Option D: When the cost of transporting fruits from South Korea to Hong Kong increases, the relative price of high-quality South Korean fruits decreases. According to the law of demand, Hong Kong will import more high-quality South Korean fruits relative to low-quality South Korean fruits. The proportion of high-quality fruits imported from South Korea increases, so the average quality of fruits imported from South Korea will increase.

Option B: When both long trips and short trips involve the same costs of quarantine, the price of long trips relative to short trips decreases. The proportion of tourists taking long trips after quarantine increases, thus increasing their average number of stay days after quarantine.

Option C: When the government increases the per-unit tax on tobacco products, the price of tobacco products increases, the quantity of tobacco products (including high-quality tobacco products) sold decreases.

16. Answer: D

The price of sending heavier parcels relative to sending lighter parcels will decrease. This will lead to an increase in the proportion of people sending heavier parcels, causing the average weight of each parcel to increase.

Options A and C: After the courier company increases its fare, its orders will decrease. The change in total expenditure will thus be uncertain.

Option B: An increase in the fare of courier service will cause the quantity demanded to decrease, demand for the courier service will remain unchanged.

17. Answer: C

Decreasing birth rate will lead to a decrease in the demand for infant formula milk powder, shifting the demand curve to the left. More manufacturers entering the infant formula milk powder market will lead to an increase in supply, shifting the supply curve to the right.

18. Answer: D

Laptop computers and tablet computers are substitutes. When the price of tablet computers decreases, the demand for laptop computers will decrease. The equilibrium position in the laptop computer market will change from A to B.

Option A: This will cause the supply of laptop computers to increase, causing the entire supply curve to shift to the right.

Option B: This will cause the supply of laptop computers to decrease, causing the entire supply curve to shift to the left.

Option C: This will cause the demand for laptop computers to increase, changing the equilibrium position in the laptop computer market from B to A.

19. Answer: C

The demand for bicycles increases. Given an upward-sloping supply curve, both the equilibrium price and quantity will increase. As the percentage change in quantity supplied (50%) is greater than the percentage change in price (10%), we can conclude that the price elasticity of supply is elastic.

20. Answer: C

Before the imposition of the tax, the price is \$16 and the quantity transacted is 11 units.

After the imposition of the tax, the price is \$18 and the quantity transacted is 10 units.

Option A: Consumers' tax burden = $(\$18 - \$16) \times 10 \text{ units} = \20

Producers' tax burden = $[\$16 - (\$18 - \$4)] \times 10 \text{ units} = \20

Consumers bear the same tax burden as producers.

Option B: Change in total expenditure = $\$18 \times 10 \text{ units} - \$16 \times 11 \text{ units} = \4 . Total expenditure will increase by \$4.

Option C: Change in total revenue net of tax = $(\$18 - \$4) \times 10 \text{ units} - \$16 \times 11 \text{ units} = -\36 .

Option D: Total government tax revenue = $\$4 \times 10 \text{ units} = \40 .

21. Answer: D

22. Answer: C

23. Answer: A

24. Answer: D

When the per-unit tax of the good is lowered from t_1 to t_2 , the price will decrease and the quantity transacted will increase. The deadweight loss will decrease by area abfg (from area acg to area bcf).

25. Answer: D

As producers sell organic grapes at a lower price while the quantity transacted remains unchanged, producer surplus will decrease.

Option A: Total expenditure will decrease as price decreases and the quantity transacted remains unchanged.

Option C: An effective price ceiling would result in a shortage, not a surplus.

26. Answer: C

When the government provides a per-unit subsidy to hotels, the supply of hotel rooms will increase. The demand for hotel rooms will increase as staying in local hotels for vacation becomes popular.

Without the government subsidy, there is no deadweight loss. Providing a per-unit subsidy will result in a deadweight loss.

Option A: As both supply and demand increase, the quantity transacted will increase while the change in the market price will be uncertain. The change in consumers' total expenditure on local hotel rooms will thus be uncertain.

Option B: As both supply and demand increase, the change in the market price will be uncertain.

Option D: Both the provision of a per-unit subsidy and an increase in demand would increase the producer surplus.

27. Answer: D

(1): As the demand for paper decreases, the quantity transacted will decrease. If the quantity transacted falls below the quota limit, the quota will become ineffective.

(2): The demand for paper decreases. If the quota is still effective, the price will decrease and the quantity transacted will remain unchanged. If the quota becomes ineffective, the price and the quantity transacted will decrease. Producer surplus will decrease in both cases.

(3): A decrease in demand reduces the gap between marginal benefit and marginal cost. The deadweight loss will decrease.

28. Answer: B

The compensation would increase the marginal private cost of making noise. The noise generated from Mr. Cheung's home will be reduced. The gap between the new marginal private cost (including the compensation) and the marginal social cost will be narrowed.

29. Answer: C

The Lorenz curve of Country B is closer to the line of perfect income inequality, so the income gap of Country B is wider than that of Country A.

Option A: The Lorenz curve of Country A is closer to the line of perfect income equality, so the income distribution of Country A is more even than that of Country B.

Option B: The Lorenz curve does not reflect the amount of income.

Option D: The Lorenz curve of Country B is closer to the line of perfect income inequality, so the Gini coefficient of Country B is larger than that of Country A.

30. Answer: B

Paper 2

Section A (30 marks)

1. (a) The scheme will increase the occupational mobility of labour. The scheme will upgrade people's vocational skills, thus increasing their ability to change from one occupation to another. (2 marks)
- (b) No, because these courses are produced from scarce resources which have alternative uses / more of these courses are preferred. (2 marks)
2. (a) Listed company / public limited company. (1 mark)
- Features:
- a legal entity
 - limited liability
 - lasting continuity
 - higher profits tax rate than that for sole proprietorships and partnerships
 - separation of ownership and management
 - more complicated set-up procedures
 - wider sources of capital than sole proprietorships and partnerships
 - shares are freely transferable
 - need to disclose its accounting information to the public
 - any reasonable answers
- (Mark the **FIRST TWO** points only)
- (b) Possible advantages:
- The company can choose not to disclose its accounting information to the public.
 - The company can reduce the risk of being taken over.
 - Any reasonable answers
- (Mark the **FIRST TWO** points only)
- (c) A way of raising capital:
- bank loans
 - retained profit
 - any reasonable answer
- (Mark the **FIRST** point only)
3. (a) Firm A's average labour productivity = $10,000 \text{ units of output} / 1,500 \text{ man-hours} = 6.67 \text{ units of output/man-hour}$ (1 mark)
- Firm B's average labour productivity = $7,500 \text{ units of output} / 1,000 \text{ man-hours} = 7.5 \text{ units of output/man-hour}$ (1 mark)
- Therefore, firm B has a higher average labour productivity. (1 mark)
- (b) Since the output quality of firm A is better than that of firm B, it shows that the workers of firm A may not need to rush for more output to earn more income. Therefore, firm A may pay its workers at time rate while firm B may pay its worker at piece rate. (3 marks)
- OR
- Any reasonable answer (3 marks)

4.

Machines (units)	Labour (units)	Total product (units)	Marginal product (units)
1	10	25	25
2	10	51	26
3	10	79	28
4	10	95	16
5	10	110	15

The law of diminishing marginal returns states that in the short run, as more units of a variable factor are added to a given quantity of fixed factor successively, marginal product will decrease eventually, other things being constant. (3 marks)

Yes. When machines increases from 3 to 4 units, the marginal product of the machines decreases from 28 to 16 units, so the law of diminishing marginal returns holds. (3 marks)

5. The behaviours of the dogs (such as barking) may disturb other people, but these people are not compensated by the dog owners (financially or otherwise). Keeping dogs may involve negative externality (external cost is involved). (3 marks)

OR

These dogs may benefit people by bringing happiness to them, but the dog owners do not receive compensation for such benefit. Keeping dogs may involve positive externality (external benefit is involved). (3 marks)

OR

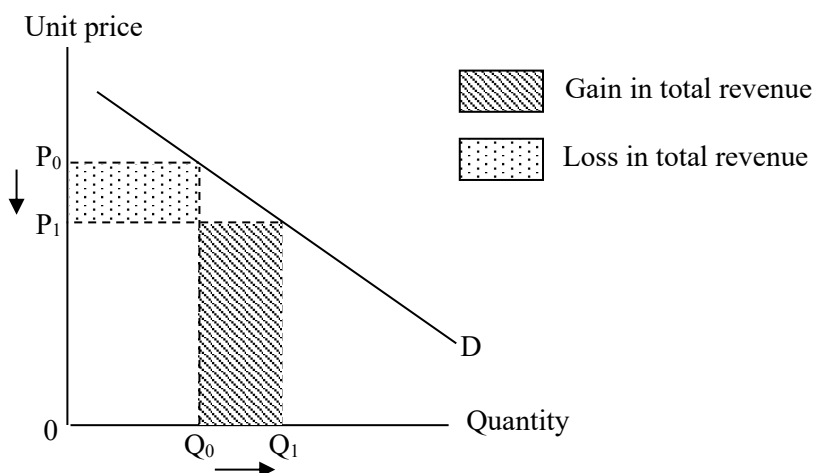
Any reasonable answer (3 marks)

6. Verbal elaboration:

If the demand for Mr. Cheung's flowers is elastic, the percentage decrease in price will be smaller than the percentage increase in quantity demanded. The total revenue for Mr. Cheung will increase. (3 marks)

Illustrate in the diagram:

- correct positions of gain in total revenue and loss in total revenue (1 mark)
- gain in total revenue > loss in total revenue (1 mark)



Section B (40 marks)

7. (a) Tertiary production, (1 mark)
Because they provide catering service. (1 mark)
- (b) No. (1 mark)
The value of their highest-valued option forgone (such as the job they have given up) may not be the same. (2 marks)
- (c) Variable cost. (1 mark)
The cost of hiring them would change as the opening hours of the restaurant changes. (1 mark)
- (d) A reason:
- The cost of measuring the output of waiters is too high.
 - The work of waiters is not standardised.
 - Any reasonable answer

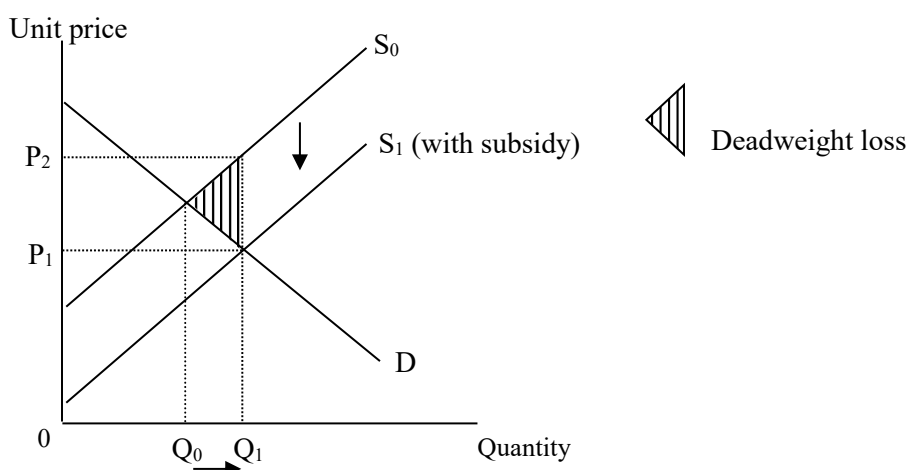
2@, max: 2 marks

(Mark the **FIRST** point only)

8. (a) After providing a per-unit subsidy on the agricultural products, the relative price of high-quality agricultural products (in terms of low-quality agricultural products) will increase. According to the law of demand, people will buy relatively fewer high-quality agricultural products. Thus, the average quality of domestic agricultural products sold will decrease. (4 marks)
- (b) Verbal elaboration:
- The supply of domestic agricultural products will increase. (1 mark)
 - The quantity transacted will increase. (1 mark)
 - Marginal benefit (P_1) will be smaller than marginal cost (P_2), and there will be a deadweight loss. (1 mark)

Illustrate in the diagram:

- downward shift of the supply curve (1 mark)
- correct position of deadweight loss (1 mark)



9. (a) The Gini coefficient based on original monthly household income increased between 2011 and 2016, reflecting that the income distribution became more uneven. (1 mark)
Since the Gini coefficient based on post-tax post-social transfer monthly household income is lower than the Gini coefficient based on original monthly household income, it shows that the government's taxes and social transfers were effective in reducing income inequality. (1 mark)

(b) A possible reason:

- Employees aged 55 or over are likely to work fewer hours than those aged 25-54, so employees aged 55 or over have a lower income.

2@, max: 2 marks

- Any reasonable answer

(Mark the **FIRST** point only)

(c) The \$2 Scheme allows the elderly to travel on designated public transport modes and services at a concessionary fare of \$2 per trip, so their opportunity cost (money cost) of going out decreases.

According to the law of demand, the elderly will go out more. (2 marks)

(d) Verbal elaboration:

Residents' services and the designated public transport modes and services are substitutes. (1 mark)

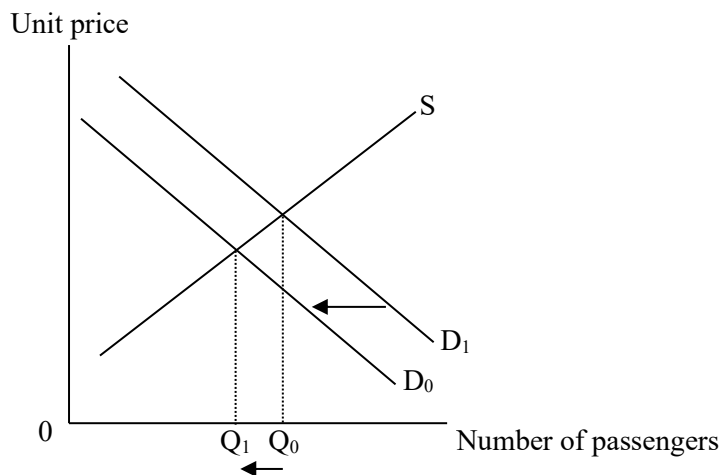
More people can travel on designated public transport modes and services at a lower price, causing the demand for residents' services to decrease. (1 mark)

The number of passengers of residents' services will decrease. (1 mark)

Illustrate in the diagram:

- leftward shift of the demand curve (1 mark)

- a decrease in the number of users (1 mark)



(e) Marks award for effective communication (EC: max 2 marks)

Marks	Performance
2	● Arguments supported with the source/data and appropriate economic theories ● Relevant materials presented ● Well-organised and coherent answers presented without repetition of ideas ● Ideas clearly and fluently expressed with appropriate use of language/words/terms/symbols
1	● Arguments presented with some support of the source/data and economic theories ● Some irrelevant materials presented ● Answers presented in a less organised way with some repetition of ideas ● A clear message conveyed with some inappropriate use of language/words/terms/symbols
0	● Arguments presented with no support of the source/data and economic theories ● Materials unrelated to the gist of the question presented ● Inconsistent arguments presented ● Limited ideas expressed with inappropriate use of words/terms/symbols

The maximum marks for content are 9 marks. Answers may include the following:

Efficiency

- Under the \$2 Scheme, eligible persons will use these public transport modes and services until marginal benefit equals the price (\$2). If the marginal cost of providing these public transport services is higher than \$2, there will be over-production, leading to a deadweight loss. Lowering the eligible age of the \$2 Scheme will increase the number of passengers taking these public transport modes and services. The output will be farther away from the efficient output level, further increasing the deadweight loss.

Gini coefficient based on post-tax post-social transfer monthly household income

- Since the elderly and the disabled mostly belong to the low-income group, lowering the eligible age of the \$2 Scheme will increase the income of the low-income group, while the income of the high-income group will remain unchanged. The income gap between the high-income group and the low-income group will be narrowed. The Gini coefficient based on post-tax post-social transfer monthly household income will decrease.

- Recipients of the \$2 Scheme can reduce their home-to-work transport costs. Lowering the eligible age of the scheme may encourage more people to go to work. The income of these people will increase. As the income of the high-income group remains unchanged, the Gini coefficient calculated based on post-tax post-social transfer monthly household income will decrease.

Other economic variables

- Mobility of labour: The \$2 Scheme can reduce the home-to-work transport costs of the beneficiaries. Lowering the eligible age of the scheme allows more people to enjoy a lower home-to-work transport cost. The geographical mobility of labour will increase.
- Labour supply: The \$2 Scheme can reduce the home-to-work transport costs of the beneficiaries. Lowering the eligible age of the scheme may encourage more people to rejoin the labour market, and the labour supply may increase.